

Supplementary Material

CreepPFN: Forecasting Concrete Creep with a Transformer Trained on a Hierarchical Bayesian Prior

Sompote Youwai, Warayut Dokduea and Weerachart Tangchirapat

This document gives the notation, the equations, the settings and diagnostics of the hierarchical Bayesian model, the network architecture, the training budgets, the data preparation, additional results and the baseline settings. All numbers are generated from the saved experiment outputs.

S1 Notation

Bold lowercase symbols denote vectors. Bold uppercase symbols denote matrices.

Symbol	Definition and unit or role
Observed data, descriptors and indices	
i, j, k, m	Curve or task index; reading or query index; basis-component index; ensemble-member index.
t, t_j, t_*	Elapsed time, time of reading j , and requested forecast time (days since loading).
$t_{a,i}$	First-reading elapsed time of curve i (days).
$J_i^{\text{obs}}(t)$	Recorded compliance of curve i at time t ($\mu\epsilon/\text{MPa}$).
$y_i(t), y_j, y_{ij}$	Compliance increment from the first reading; context or target increment at reading j ($\mu\epsilon/\text{MPa}$).
$\hat{y}$	Predicted response median ($\mu\epsilon/\text{MPa}$).
ρ	Concrete density (kg/m^3).
f_c	Recorded compressive strength (MPa).
E_{28}	Recorded 28-day elastic modulus (MPa).
E_{rec}	External-cohort recorded modulus, used as an unverified proxy for E_{28} (MPa).
r_σ	Recorded ratio of applied stress to compressive strength at loading (dimensionless).
$\mathcal{C}, n_c$	Set of observed context time–response pairs and its number of readings.
Q_i, N	Query-index set of task i and number of tasks in a mini-batch.
Hierarchical Bayesian prior	
$g(t)$	Kelvin4 compliance increment of one curve ($\mu\epsilon/\text{MPa}$).
τ_k	Retardation time of Kelvin component k : 0.3, 3 and 30 days.
$\phi_k(t), h_k(t; t_a)$	Raw basis function and its anchor- and day-160-normalized form.
A_k, M	Nonnegative amplitude of component k and day-160 magnitude $M = \sum_k A_k$ ($\mu\epsilon/\text{MPa}$).
$\mathbf{x}_i$	Training-standardized seven-channel descriptor vector.
$\mathbf{z}_i$	Curve parameters on the unconstrained scale: $\log M$ and three log amplitude ratios.
$\beta_0, \mathbf{B}$	Regression intercept vector and descriptor-to-parameter coefficient matrix.
$s(i), \mathbf{u}_s$	Literature source of curve i and its source effect.
τ_u, τ_e	Standard deviations of source and specimen effects.
e_{ij}, σ_i, r_i	Reading error, its standard deviation and its ratio to M_i .
μ_r, s_r	Mean and standard deviation of $\log r_i$.
$D_i(t), W, \ell, \kappa$	Discrepancy path, standard Brownian motion, log-time since the first reading and discrepancy scale.
$\hat{R}$	Split potential scale reduction factor, a Markov chain convergence diagnostic.
Network and predictive distribution	
s	Context-only response scale, $\max(\max_j y_j , 1 \mu\epsilon/\text{MPa})$ ($\mu\epsilon/\text{MPa}$).
$\mathbf{c}_j, \mathbf{d}_*$	Three-feature context token and four-feature query token before projection.

Symbol	Definition and unit or role (continued)
$P_{\text{prop}}, P_c, P_q$	Learned property, context-token and query-token projection networks.
Enc_4, LN	Four-layer context encoder and layer normalization.
Attn, Head	Query-to-context attention operation and predictive output head.
$\mathbf{p}, H, \mathbf{q}_*$	Property embedding, encoded context matrix and projected query embedding.
r_*, a_*	Predicted location residual and unconstrained scale output for one query.
z	Transformed response $\text{asinh}(y/s)$.
μ, σ	Gaussian location and standard deviation on the transformed response scale.
$\mathcal{L}$	Task-balanced Gaussian negative log-likelihood training objective.
Evaluation	
$\text{MAE}_i, \text{NMAE}_i$	Mean absolute error of curve i ($\mu\epsilon/\text{MPa}$) and its normalized form.
CRPS	Continuous ranked probability score, a proper score of a predictive distribution.

S2 Equations

S2.1 Target, error and coverage

The target is the compliance increase after the first reading,

$$y_i(t) = J_i(t) - J_i(t_{a,i}), \quad (\text{S1})$$

where J_i is the recorded compliance of curve i in $\mu\epsilon/\text{MPa}$ and $t_{a,i}$ is its first-reading time. For curve i with later-reading set Q_i ,

$$\text{MAE}_i = \frac{1}{|Q_i|} \sum_{j \in Q_i} |\hat{y}_{ij} - y_{ij}|, \quad \text{NMAE}_i = \frac{\text{MAE}_i}{\max\left(|Q_i|^{-1} \sum_{j \in Q_i} |y_{ij}|, 1 \mu\epsilon/\text{MPa}\right)}. \quad (\text{S2})$$

The reported NMAE averages NMAE_i within each literature source and then over sources with equal weight. Coverage is

$$C_i = \frac{1}{|Q_i|} \sum_{j \in Q_i} \mathbf{1}(L_{ij} \leq y_{ij} \leq U_{ij}), \quad (\text{S3})$$

where $[L_{ij}, U_{ij}]$ is the 90% range, averaged in the same way. Confidence intervals are percentiles of 2,000 bootstrap resamples of source averages (Efron and Tibshirani, 1994). Paired differences resample the same sources for both methods.

S2.2 Hierarchical Bayesian model

For curve i with first-reading time t_a , the Kelvin4 family is

$$g(t) = \sum_{k=1}^4 A_k h_k(t; t_a), \quad h_k(t; t_a) = \frac{\phi_k(t) - \phi_k(t_a)}{\phi_k(160) - \phi_k(t_a)}, \quad (\text{S4})$$

with $\phi_k(t) = 1 - \exp(-t/\tau_k)$ for $\tau_k \in \{0.3, 3, 30\}$ days, $\phi_4(t) = \log(1 + t/10)$, $A_k \geq 0$ and day-160 size $M = \sum_k A_k$. The unconstrained parameters are $\mathbf{z} = (\log M, \log A_1/A_4, \log A_2/A_4, \log A_3/A_4)$. The hierarchical model is

$$\mathbf{z}_i \sim \mathcal{N}(\boldsymbol{\beta}_0 + \mathbf{x}_i^\top \mathbf{B} + \mathbf{u}_{s(i)}, \text{diag } \boldsymbol{\tau}_e^2), \quad \mathbf{u}_s \sim \mathcal{N}(\mathbf{0}, \text{diag } \boldsymbol{\tau}_u^2), \quad (\text{S5})$$

$$y_{ij} = g(t_{ij}; \mathbf{z}_i) + e_{ij} - e_{i0} + D_i(t_{ij}), \quad e_{ij} \sim \mathcal{N}(0, \sigma_i^2), \quad \sigma_i = M_i r_i, \quad \log r_i \sim \mathcal{N}(\mu_r, s_r^2). \quad (\text{S6})$$

Here $\mathbf{x}_i$ is the standardized seven-descriptor vector and $s(i)$ is the literature source of curve i . The first-reading error e_{i0} is subtracted because the response is measured from that reading. The discrepancy is

$$D_i(t) = \kappa M_i W(\ell(t)), \quad \ell(t) = \log(1 + t) - \log(1 + t_a), \quad (\text{S7})$$

where W is a standard Brownian motion that starts at zero at t_a . Its covariance is $\kappa^2 M_i^2 \min(\ell_j, \ell_k)$, so it grows with log-time since the first reading. The intercept has a normal prior with standard deviation 2 centred on the median least-squares fit. The coefficients $\mathbf{B}$ have normal priors with standard deviation 0.5, the spreads have half-normal priors with scale 1, $\mu_r \sim \mathcal{N}(\log 0.015, 1)$ and $s_r \sim \text{half-normal}(0.5)$.

S2.3 Network

For the context readings $\mathcal{C} = \{(t_j, y_j) : j \leq n_c\}$, the descriptors $\mathbf{x}$ and a requested day t_* , the network uses the context scale

$$s = \max\left(\max_{j \leq n_c} |y_j|, 1 \mu\epsilon/\text{MPa}\right). \quad (\text{S8})$$

Context tokens hold $\log(1 + t_j)/\log(161)$, $t_j/160$ and $\text{asinh}(y_j/s)$. Query tokens hold linear time, log time, the log time since the last context reading and the transformed last response. The forward map is

$$\begin{aligned} \mathbf{p} &= P_{\text{prop}}([\mathbf{x}, \log(1 + s)]), \quad H = \text{LN}(\text{Enc}_4(\{P_c(\mathbf{c}_j) + \mathbf{p}\}_{j \leq n_c})), \\ \mathbf{q}_* &= P_q(\mathbf{d}_*) + \mathbf{p}, \quad (r_*, a_*) = \text{Head}[\mathbf{q}_* + \text{Attn}(\mathbf{q}_*, H, H)], \\ \mu_* &= \text{asinh}(y_{n_c}/s) + r_*, \quad \sigma_* = \text{softplus}(a_*) + 0.03. \end{aligned} \quad (\text{S9})$$

P_{prop} is the property network, P_c and P_q are the context and query projections, Enc_4 is a four-layer pre-normalized transformer encoder (Vaswani et al., 2017) with GELU activation (Hendrycks and Gimpel, 2016), and Attn is cross-attention. A padding mask excludes absent readings. Queries share the encoded context and do not exchange information. No positional encoding is used; readings are distinguished by the time values in each token.

S2.4 Training objective and ranges

For $z = \text{asinh}(y/s)$, training minimizes the task-balanced Gaussian negative log-likelihood

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \frac{1}{|Q_i|} \sum_{j \in Q_i} \left[\frac{(z_{ij} - \mu_{ij})^2}{2\sigma_{ij}^2} + \log \sigma_{ij} + \frac{1}{2} \log(2\pi) \right]. \quad (\text{S10})$$

A network's median and 90% range are $\hat{y} = s \sinh(\mu)$ and $[L, U] = s \sinh(\mu \pm 1.6448536\sigma)$. The three-network forecast is $s \sum_{m=1}^3 \sinh(\mu_m)/3$. Its range uses the equal-weight mixture

$$F(y) = \frac{1}{3} \sum_{m=1}^3 \Phi\left(\frac{\text{asinh}(y/s) - \mu_m}{\sigma_m}\right), \quad [L, U] = [F^{-1}(0.05), F^{-1}(0.95)], \quad (\text{S11})$$

solved by bisection.

S3 Hierarchical Bayesian model: settings and diagnostics

Each fold fits its own model to its training sources: 365–402 curves from 41–42 sources. The parameters were fitted with $\kappa = 0$ by the No-U-Turn Sampler (Hoffman and Gelman, 2014) in NumPyro (Phan et al., 2019), with four chains of 1,500 warm-up and 500 retained draws, a target acceptance rate of 0.9 and a maximum tree depth of 10. Convergence was assessed with the split $\hat{R}$ and the bulk effective sample size (Vehtari et al., 2021). The likelihood uses all readings with the anchored residual covariance $\sigma_i^2(\mathbf{I} + \mathbf{1}\mathbf{1}^\top)$. Chains start at the least-squares fits. Table S2 reports the diagnostics. A version with an additional 300-day term reached maximum $\hat{R}$ of 1.07–1.82 because that term and the logarithmic term are nearly collinear over 160 days (Figure S1a). The curve family and the discrepancy scale κ were then chosen on validation sources. For each candidate, the forecasts implied by the model were computed by Monte Carlo sampling from the readings up to day 10 and scored by the continuous ranked probability score (CRPS), a proper score of accuracy and uncertainty (Gneiting and Raftery, 2007) (Table S3). Kelvin4 with $\kappa = 0.01$ scored best and was used to generate all practice tests.

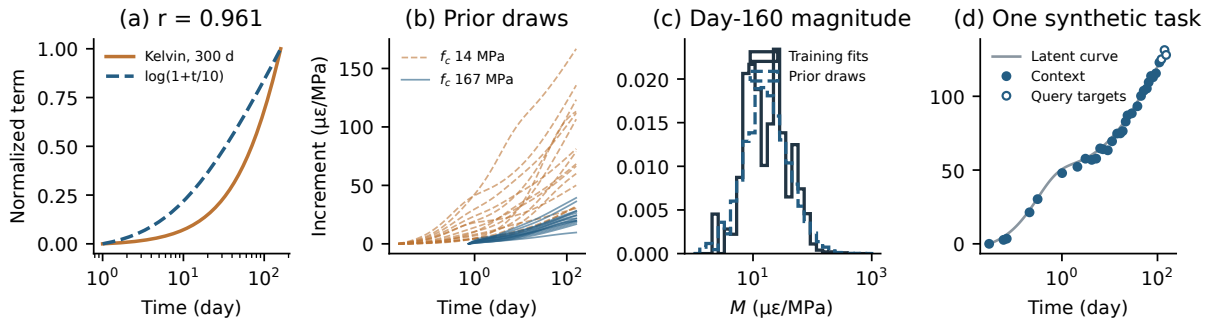
Table S4 follows three fold-1 training curves through the model. The fitted shapes vary widely: the 90th-percentile curve is almost purely logarithmic, whereas the median curve has most of its increase in the short retardation terms. The descriptor-only prediction of M ranges only from 28.6 to 45.1 $\mu\epsilon/\text{MPa}$, while the fitted sizes range from 10.7 to 82.2 $\mu\epsilon/\text{MPa}$. Source and specimen effects supply the rest. Table S5 quantifies this for all folds.

Table S2: Sampler diagnostics per fold. Max $\hat{R}$ is the largest split $\hat{R}$ over global parameters. Min ESS is the smallest bulk effective sample size. Noise is the posterior median of $\exp(\mu_r)$.

Fold	Divergences	Max $\hat{R}$	Min ESS	Noise (% of M)	Minutes
1	0	1.033	163	2.03	21
2	2	1.052	103	2.06	21
3	0	1.042	100	2.47	19
4	1	1.065	56	2.47	21
5	0	1.048	68	2.55	20

Table S3: Validation choice of curve family and discrepancy scale. Values are source-averaged percentages on the validation sources, from readings up to day 10. The setting with the lowest normalized CRPS was selected.

Curve family	κ	NMAE (%)	Coverage (%)	CRPS (%)
Kelvin4	0.00	14.89	91.2	10.67
Kelvin4	0.01	14.82	91.8	10.62
Kelvin4	0.02	14.95	92.6	10.69
Kelvin4	0.03	15.41	91.7	11.01
Kelvin4	0.05	16.55	89.9	11.75
Weibull + Kelvin4	0.00	15.17	89.1	10.93
Weibull + Kelvin4	0.01	15.10	90.3	10.88
Weibull + Kelvin4	0.02	15.20	91.4	10.87
Weibull + Kelvin4	0.03	15.65	91.1	11.15
Weibull + Kelvin4	0.05	16.74	88.5	11.83

**Figure S1:** The hierarchical model for fold 1. (a) The 300-day Kelvin term and the logarithmic term, normalized over 1–160 days. (b) Fifteen draws each for the training records with the lowest and highest strength among donors drawn at least 15 times. (c) Day-160 size M of 4,000 draws and of least-squares Kelvin4 fits to the training curves. (d) One practice test with input readings and hidden targets.**Table S4:** Three fold-1 training curves at the 10th, 50th and 90th percentiles of the least-squares Kelvin4 size M . Modulus is in GPa. Shares are A_k/M in percent for the 0.3-, 3- and 30-day terms and the logarithmic term. The last column is the descriptor-only prediction of M at the posterior mean. M is in $\mu\epsilon/\text{MPa}$.

Pct.	Curve	Source	ρ	f_c	E_{28}	r_σ	Fitted M	Shares (%)	Predicted M
10	NU00634	NU_L367	2420	67.2	40.6	0.30	10.7	28 / 8 / 10 / 54	28.6
50	NU00600	NU_L367	2380	43.5	35.1	0.30	34.0	30 / 31 / 12 / 27	45.1
90	NU00361	NU_L95	2558	50.1	29.8	0.14	82.2	8 / 0 / 0 / 92	38.0

Table S5: Decomposition of $\log M$ in each fold’s model at posterior means. SD desc. is the standard deviation of the descriptor prediction over the training records. τ_u and τ_e are the source and specimen standard deviations. Shares are percentages of the total variance. R^2 compares the descriptor prediction with least-squares $\log M$.

Fold	Curves	SD desc.	τ_u	τ_e	Desc. (%)	Source (%)	Specimen (%)	R^2
1	402	0.45	0.55	0.50	26.6	40.4	33.0	0.27
2	365	0.45	0.57	0.49	26.5	42.3	31.1	0.16
3	395	0.46	0.51	0.51	28.7	35.5	35.8	0.28
4	382	0.47	0.49	0.46	32.8	35.4	31.8	0.35
5	382	0.44	0.49	0.45	30.6	37.4	32.0	0.36

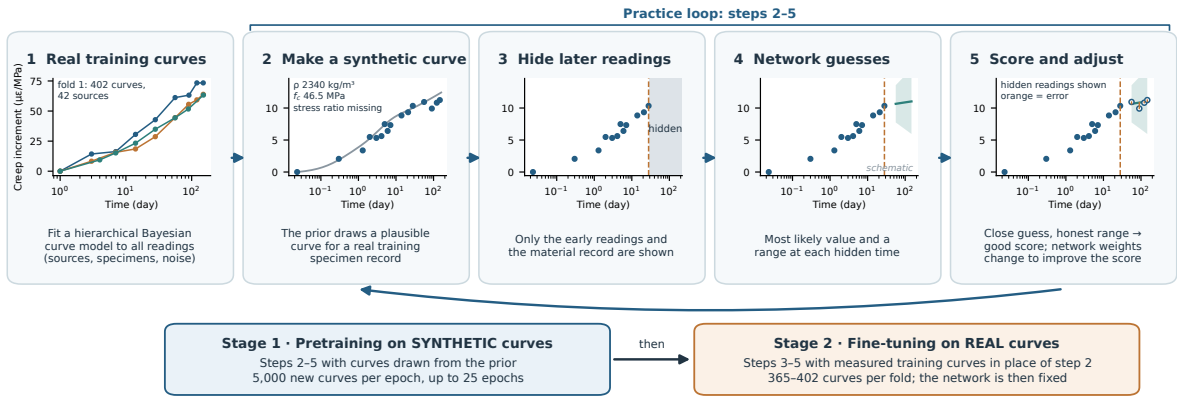
S4 Network architecture

Table S6 lists the network components. Each network has width 256, four encoder layers, four attention heads and feed-forward width 768, with 3,100,418 trainable weights. Figure S2 illustrates the training loop and a single forecast.

Table S6: Network components and parameter counts. Dropout is 0.05 in the encoder and cross-attention, except in fold 3, where it is zero.

Component	Operation	Parameters
Context projection	Linear, $3 \rightarrow 256$	1,024
Property network	Linear $8 \rightarrow 256$, GELU, linear $256 \rightarrow 256$	68,096
Context encoder	4 pre-norm transformer layers; each has layer norm, 4-head self-attention, layer norm and feed-forward $256 \rightarrow 768 \rightarrow 256$ with GELU; padding mask	2,633,728
Context normalization	Layer norm	512
Query projection	Linear $4 \rightarrow 256$, GELU, linear $256 \rightarrow 256$	67,072
Cross-attention	1 layer, 4 heads; queries from query tokens, keys and values from encoded context	263,168
Query residual path	Adds the query token to the attention output	0
Output head	Layer norm, linear $256 \rightarrow 256$, GELU, linear $256 \rightarrow 2$; last layer starts at zero weights and bias $(0, -1)$	66,818
Total per member	Three members per fold	3,100,418

A Training: done once per fold, using training sources only



B Application: forecast a new specimen (NU00017, from a withheld test source)

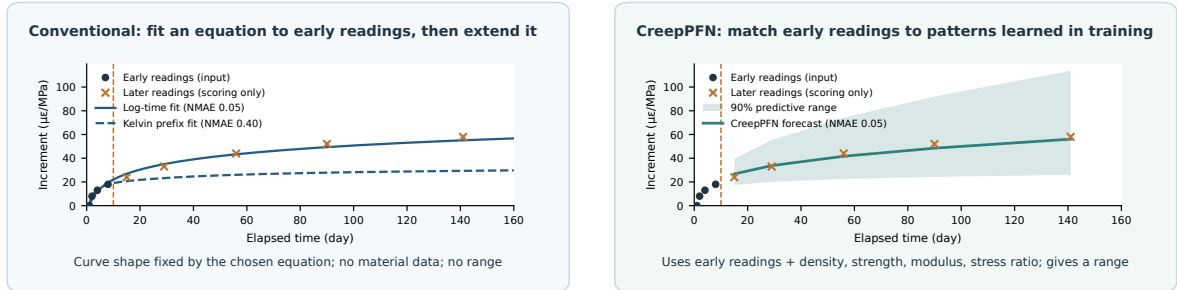


Figure S2: Training loop and one forecast. (A) The practice loop in fold 1: measured training curves, one practice test drawn from the fold-1 model, hidden readings, and a schematic guess and score. (B) Test curve NU00017 forecast from its readings up to day 10 by the logarithmic and Kelvin fits (left) and by CreepPFN (right).

S5 Training

Each practice round draws 5,000 new tests in blocks of eight. A block shares one training source, one posterior draw of the global parameters and one source effect. Each test copies the descriptors and reading times of a training curve from that source, draws a specimen effect, a noise level and a discrepancy path, and splits its readings at a random point into input and hidden readings. Pretraining uses AdamW (Loshchilov and Hutter, 2019) with weight decay 10^{-3} , batch size 128, cosine learning-rate decay to one tenth of the initial rate over at most 25 rounds, gradient clipping at one and early stopping after six rounds without validation improvement. The learning rate is 6×10^{-4} in folds 1–2 and 3×10^{-4} in folds 3–5. Fine-tuning uses the measured training curves with source-balanced sampling, learning rate 10^{-5} , batch size 64, at most 20 epochs and five-epoch patience. The pretrained checkpoint is kept if fine-tuning does not improve validation error. Three networks with seeds 45, 46 and 47 are trained per fold (Table S7). Median pretraining time was 58 s per network on one GPU.

Table S7: Training budgets per fold for seeds 45 / 46 / 47. “Tasks to checkpoint” is in thousands. A fine-tuning best epoch of 0 means the pretrained checkpoint was kept.

Fold	Rounds run	Best round	Tasks to checkpoint (k)	Fine-tuning best epoch	Improved
1	13 / 14 / 11	7 / 8 / 5	35 / 40 / 25	9 / 0 / 0	1 of 3
2	21 / 17 / 10	15 / 11 / 4	75 / 55 / 20	7 / 8 / 7	3 of 3
3	18 / 17 / 11	12 / 11 / 5	60 / 55 / 25	0 / 0 / 0	0 of 3
4	19 / 18 / 25	13 / 12 / 20	65 / 60 / 100	1 / 0 / 2	2 of 3
5	25 / 11 / 16	24 / 5 / 10	120 / 25 / 50	0 / 1 / 6	2 of 3

S6 Data

The NU extract contains 617 curves from 65 literature sources and 9,790 readings up to day 160 (Hubler et al., 2015). Figure S3 summarizes the NU and external curves. Flagged readings were removed and duplicate times averaged. Import required at least four readings, a first reading by day 10, a last reading at or after day 28, a source identifier and positive density and strength. The inherited extract also applied descriptor, duration and fit-quality filters, which can favour smoother or better-recorded experiments. Recorded E_{28} is missing for 213 of 617 curves, and 24 sources have no recorded modulus. Among the 610 evaluated curves, 426 have a recorded stress ratio (Figure S4). Source sizes range from one to 58 curves.

The external workbook contains 66 curves and 1,195 readings, with 350 readings up to day 10 and 845 later readings up to day 156. Strain is in microstrain, and strength and modulus were converted from kgf/cm² to MPa. A nominal stress of $0.4f_c$ converts strain to compliance, following the local data convention. The stress ratio is treated as missing, and the recorded modulus is an unverified proxy for E_{28} . The external curves were checked against the KMUTT provenance workbook, which lists the laboratory mixes with their properties and readings. Curves PA000–PA046 match laboratory mixes exactly and form the KMUTT set of 47 curves. An earlier compilation had assigned PA041–PA046 to the literature set by their row position alone. The remaining 19 curves, PA047–PA065, are compiled from the five published studies listed by the earlier compilation (Rossi et al., 2013, Choi et al., 2025, Hong et al., 2024, Zhao et al., 2016, Gao et al., 2025). The density, strength and modulus of PA057–PA060 match the four mixtures of Hong et al. (Hong et al., 2024), and PA061 matches the 28-day modulus of the reference concrete of Zhao et al. (Zhao et al., 2016). The curves PA047–PA056 are ten normal-weight mixes with strengths of 26–37 MPa. They cannot come from Rossi et al. (Rossi et al., 2013), who tested one concrete at three stress levels, nor from Choi et al. (Choi et al., 2025), who tested lightweight concrete. By elimination, they most likely come from the recycled-aggregate-concrete compilation of Gao et al. (Gao et al., 2025). These matches are probable but not confirmed against the original curves. The external errors differ strongly between these blocks. Averaged over the five fold models, the NMAE was 31.6% for PA047–PA056 (range 20.2–37.4%), 11.0% for PA057–PA060 and 11.2% for PA061–PA065. The late-to-early creep-rate ratio, computed from the later readings of each curve, had a median of 0.49 for the literature curves and 0.30 for the KMUTT curves. The curves PA047–PA056 therefore keep rising more strongly than basic creep curves.

Curve identifiers were sorted before five-fold grouped splitting by literature source (Table S8). Within

each development partition, about 20% of sources formed the validation set. The training sources alone fitted the Bayesian model, the descriptor scaler and the imputation medians.

Table S8: Fold composition as curves/sources. Eligible test curves have at least two readings up to day 10 and one later reading.

Fold	Train	Validation	Test	Eligible test curves	Later readings
1	402/42	91/11	124/12	124	918
2	365/41	128/10	124/14	124	1,036
3	395/42	99/10	123/13	122	1,019
4	382/42	112/10	123/13	120	1,648
5	382/42	112/10	123/13	120	1,025
All test folds	—	—	617/65	610	5,646

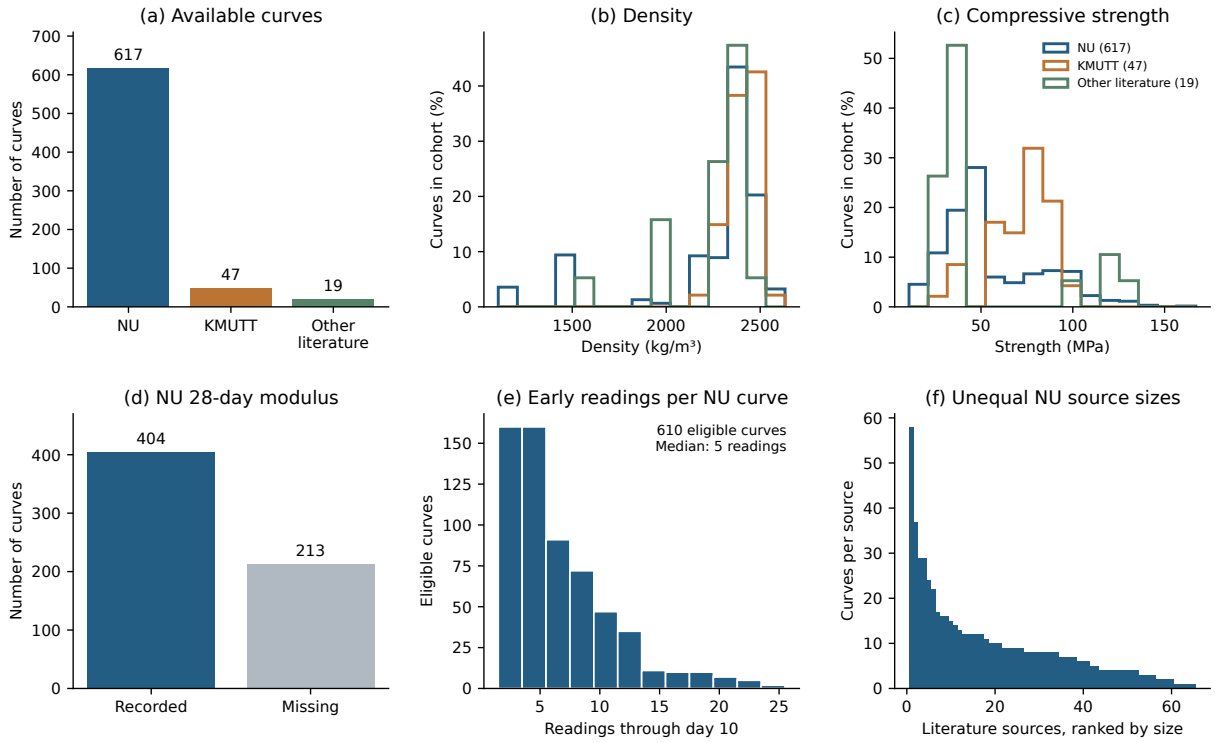


Figure S3: Data characteristics. (a)–(c) Curve counts, density and strength for NU, KMUTT and the literature set. (d)–(f) NU modulus availability, readings up to day 10 for the 610 evaluated curves and the 65 source sizes.

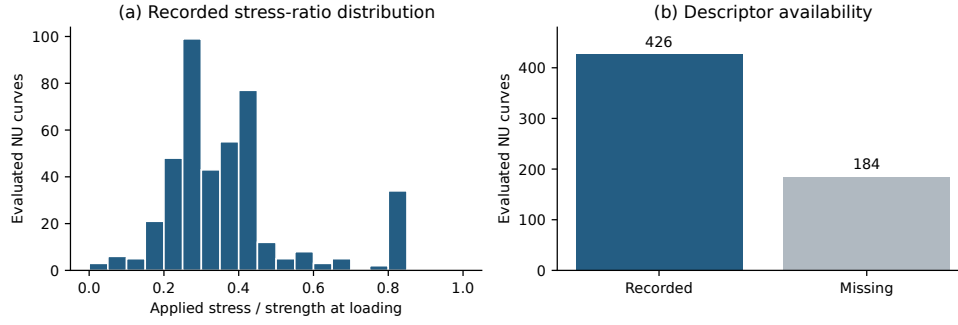


Figure S4: Recorded stress ratio and its availability among the 610 evaluated NU curves.

S7 Additional results

Table S9 reports the test results of the proposed CreepPFN, the same network before fine-tuning, the earlier CreepPFN trained on the empirical generator and the curve fits. Table S10 gives the paired contrasts. Table S11 gives coverage by forecast distance, and Table S12 gives the test-duration results. Figure S5 compares the two synthetic generators.

Table S9: Test forecasts on 610 NU curves, 65 sources and 5,646 later readings. NMAE is source-averaged, in percent, with a 95% bootstrap interval. MAE and width are in $\mu\epsilon/\text{MPa}$. The earlier CreepPFN is the same network trained on the earlier empirical generator; its variant without the query residual path omits the addition of the query token to the attention output.

Method	NMAE (%)	MAE	Coverage (%)	Width
CreepPFN (proposed)	15.94 [13.78, 18.22]	6.06	93.6	30.4
CreepPFN before fine-tuning	16.12 [13.90, 18.57]	6.11	94.6	32.8
Earlier CreepPFN, no query residual path	16.70 [14.08, 19.55]	6.14	93.0	28.3
Earlier CreepPFN (empirical generator)	17.12 [14.71, 19.79]	6.44	94.4	29.9
Kelvin fit of early readings	26.51 [23.58, 29.90]	9.19	—	—
Logarithmic fit of early readings	26.78 [18.29, 41.22]	10.48	—	—
Persistence	47.26 [44.37, 50.27]	17.85	—	—

Note: Bold marks the lowest NMAE and MAE, and coverage at or above the nominal 90%. Curve fits give no range, so their coverage and width are not defined (—). A narrower width is better only when coverage is at the nominal level, so width is not bolded.

Table S10: Paired source contrasts in percentage points of NMAE (first minus second), with 95% intervals from 2,000 paired resamples. The last column counts sources in which the first method has lower error.

Contrast	Difference	95% interval	Sources
CreepPFN (proposed) – earlier CreepPFN	-1.18	[-2.40, +0.03]	40/65
CreepPFN (proposed) – earlier CreepPFN, no query residual path	-0.76	[-2.10, +0.59]	40/65
CreepPFN before fine-tuning – earlier CreepPFN	-1.00	[-2.27, +0.24]	43/65
CreepPFN (proposed) – CreepPFN before fine-tuning	-0.18	[-0.42, +0.05]	35/65
CreepPFN (proposed) – logarithmic fit	-10.84	[-24.92, -2.67]	49/65
CreepPFN (proposed) – Kelvin fit	-10.57	[-13.58, -7.67]	54/65
CreepPFN (proposed) – persistence	-31.32	[-34.69, -27.88]	64/65

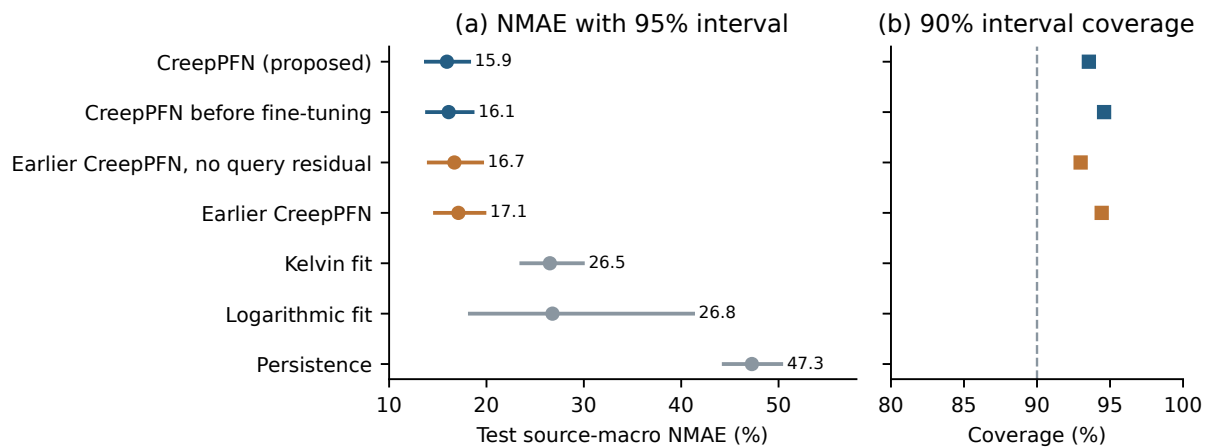
Table S11: Coverage of CreepPFN’s 90% range by distance from the last input reading, pooled over the 5,646 test readings. Relative width is the median range width divided by the observed value.

Distance (days)	Readings	Coverage (%)	Relative width
0–14	1,244	96.9	0.49
14–30	1,090	95.6	0.60
30–60	1,299	94.7	0.71
60–90	907	93.5	0.84
90–160	1,106	91.3	0.95

Table S12: Test duration on 556 test curves, all scored on readings after day 28.

Days of readings	CreepPFN NMAE (%)	Coverage (%)	Logarithmic fit NMAE (%)
3	25.40	94.2	43.88
5	22.99	95.7	33.73
7	18.64	94.2	26.23
10	17.20	93.1	22.50
14	14.80	94.4	19.25
21	11.55	93.1	16.11
28	9.58	93.7	13.99

Note: Bold marks the lower NMAE of the two methods at each test duration.

**Figure S5:** Test accuracy and coverage of CreepPFN trained on the hierarchical Bayesian model (blue), the same network trained on the earlier generator (orange) and the curve fits (grey).

S8 SHAP analysis

SHAP values follow Lundberg and Lee (Lundberg and Lee, 2017). The explained output is the mean of the three network medians, in $\mu\epsilon/\text{MPa}$, at days 28, 90 and 160. Each of the 610 test curves is explained only by the fold model that withheld its source. The early readings, their times and the first-reading time are held at the test curve’s own values in every evaluation. The four players are density; strength; modulus with its missing-value flag; and stress ratio with its missing-value flag. A descriptor and its flag are replaced together, so that no inconsistent missing-value code is created. For each fold, the background contains one eligible curve drawn uniformly from each training source. A second background is drawn independently with a different seed; 155 of the 209 background slots differ between the two. All $2^4 = 16$ coalitions are evaluated. For a coalition, the present descriptors come from the test curve and the absent descriptors from the same background curve, and the output is averaged over the background curves. Exact Shapley weights then give

$$\phi_j = \sum_{S \subseteq \mathcal{D} \setminus \{j\}} \frac{|S|! (4 - |S| - 1)!}{4!} [v(S \cup \{j\}) - v(S)], \quad (\text{S12})$$

where $\mathcal{D}$ is the set of descriptor groups and $v(S)$ is the background-averaged forecast with the groups in S and the early readings taken from the test curve. The contributions plus the reference forecast $v(\emptyset)$ reproduce each prediction to within $10^{-8} \mu\epsilon/\text{MPa}$, and the full-coalition value reproduces the model’s direct prediction to within $5 \times 10^{-5} \mu\epsilon/\text{MPa}$. Importance averages $|\phi_j|$ over curves within a source and then over sources. Direction is the Spearman correlation between a descriptor’s recorded value and its own ϕ_j at day 90. The coalition values were computed in one GPU run that also enumerated the early readings as a fifth group; the conditional game uses the 16 coalitions in which the readings are the test curve’s own. The replaced inputs can form combinations outside the observed data, and correlated inputs share credit. The run took 188 s on one RTX 3060.

S9 Baseline settings

All learners were fitted per fold on the training sources, tuned on the validation sources and scored once on the test sources. The tabular input set holds the seven descriptors, $\log(1 + t)$ and $\log(1 + t_a)$. These models were trained on every reading of the training curves. The prefix input set holds the seven descriptors, $\log(1 + t_*)$, $\log(1 + t_* - t_{nc})$, $\log(1 + t_{nc})$, $\log(1 + t_a)$, $\log(1 + s)$, y_{nc}/s , n_c , and the logarithmic and Kelvin-fit forecasts at t_* divided by s . The target is y/s . These models were trained on tasks with cutoffs of 3, 5, 7, 10, 14, 21 and 28 days from the training curves. Hyperparameters were chosen by validation NMAE from these grids. XGBoost (Chen and Guestrin, 2016): 300 or 800 trees, depth 3, 5 or 7, learning rate 0.03 or 0.1, minimum child weight 1 or 10, row and column subsampling 0.8. Random forest (Breiman, 2001): 300 trees, minimum leaf size 1, 5 or 20, feature fraction 0.33, 0.66 or 1. MLP (scikit-learn (Pedregosa et al., 2011)): hidden layers (64, 64), (128, 128) or (256, 128, 64), L2 penalty 10^{-4} or 10^{-2} , at most 600 iterations with early stopping. Inputs of the MLP are standardized. The 90% range of each learner is $\hat{y} \pm q s$, where q is the split-conformal (Angelopoulos and Bates, 2023) quantile of $|y - \hat{y}|/s$ over the n validation readings at level $0.9(1 + 1/n)$. Quantile XGBoost used the selected settings with the quantile loss at 0.05, 0.5 and 0.95. Each learner used one seed.

References

- Angelopoulos, A.N., and Bates, S. (2023). Conformal prediction: a gentle introduction. *Foundations and Trends in Machine Learning*, 16(4), 494–591. <https://doi.org/10.1561/22000000101>.
- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>.
- Chen, T., and Guestrin, C. (2016). XGBoost: a scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>.
- Choi, J.-S., Hong, S.-H., Yoo, S.-J., Yang, J.-M., and Yoon, Y.-S. (2025). Effect of internal curing on shrinkage and creep of self-compacting lightweight concrete under different curing conditions. *Construction and Building Materials*, 470, 140612. <https://doi.org/10.1016/j.conbuildmat.2025.140612>.

- Efron, B., and Tibshirani, R.J. (1994). *An Introduction to the Bootstrap*. Chapman and Hall/CRC, New York. <https://doi.org/10.1201/9780429246593>.
- Gao, C., Luo, C., Zhang, B., Hu, Z., Tang, J., Wang, L., and Liu, J. (2025). Creep behavior evaluation of recycled aggregate concrete using machine learning technology. *Journal of Building Engineering*, 99, 111538. <https://doi.org/10.1016/j.jobe.2024.111538>.
- Gneiting, T., and Raftery, A.E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. <https://doi.org/10.1198/016214506000001437>.
- Hendrycks, D., and Gimpel, K. (2016). Gaussian error linear units (GELUs). arXiv:1606.08415. <https://doi.org/10.48550/arXiv.1606.08415>.
- Hoffman, M.D., and Gelman, A. (2014). The No-U-Turn Sampler: adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15, 1593–1623. arXiv:1111.4246. <https://doi.org/10.48550/arXiv.1111.4246>.
- Hong, S.-H., Choi, J.-S., Yoo, S.-J., Yoo, D.-Y., and Yoon, Y.-S. (2024). Reinforcing effect of CNT on the microstructure and creep properties of high-strength lightweight concrete. *Construction and Building Materials*, 428, 136294. <https://doi.org/10.1016/j.conbuildmat.2024.136294>.
- Hubler, M.H., Wendner, R., and Bažant, Z.P. (2015). Comprehensive database for concrete creep and shrinkage: analysis and recommendations for testing and recording. *ACI Materials Journal*, 112, 547–558. <https://doi.org/10.14359/51687453>.
- Loshchilov, I., and Hutter, F. (2019). Decoupled weight decay regularization. *International Conference on Learning Representations*. arXiv:1711.05101. <https://doi.org/10.48550/arXiv.1711.05101>.
- Lundberg, S.M., and Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. arXiv:1705.07874. <https://doi.org/10.48550/arXiv.1705.07874>.
- Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. arXiv:1201.0490. <https://doi.org/10.48550/arXiv.1201.0490>.
- Phan, D., Pradhan, N., and Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in NumPyro. arXiv:1912.11554. <https://doi.org/10.48550/arXiv.1912.11554>.
- Rossi, P., Tailhan, J.-L., and Le Maou, F. (2013). Creep strain versus residual strain of a concrete loaded under various levels of compressive stress. *Cement and Concrete Research*, 51, 32–37. <https://doi.org/10.1016/j.cemconres.2013.04.005>.
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., and Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30, 5998–6008. arXiv:1706.03762. <https://doi.org/10.48550/arXiv.1706.03762>.
- Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: an improved $\hat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, 16(2), 667–718. <https://doi.org/10.1214/20-BA1221>.
- Zhao, Q., Yu, J., Geng, G., Jiang, J., and Liu, X. (2016). Effect of fiber types on creep behavior of concrete. *Construction and Building Materials*, 105, 416–422. <https://doi.org/10.1016/j.conbuildmat.2015.12.149>.